

EAT40006

Engineering Technology Project B

Sprint Two Report



Portfolio Task 6

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Swinburne University of Technology

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Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging. We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands. We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors. We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.

Contribution Summary (This Sprint)

All team members contributed collaboratively to Sprint 2. The following table summarises each member's contribution across key dimensions of initiative, communication, and respect.

Team Member	Tasks Done On Time	Self-Assigned	Team Meetings	Supervisor Meetings	Client Meetings	Replied Timely	Team Updated	Respected Others
Yeshan Randika Bothalage	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Rick Shevon	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sithum Vishmitha S.H.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ruzmy Mohamed	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Chiraath Madahapola	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Jonathan Anak George	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Contribution Details (This Sprint)

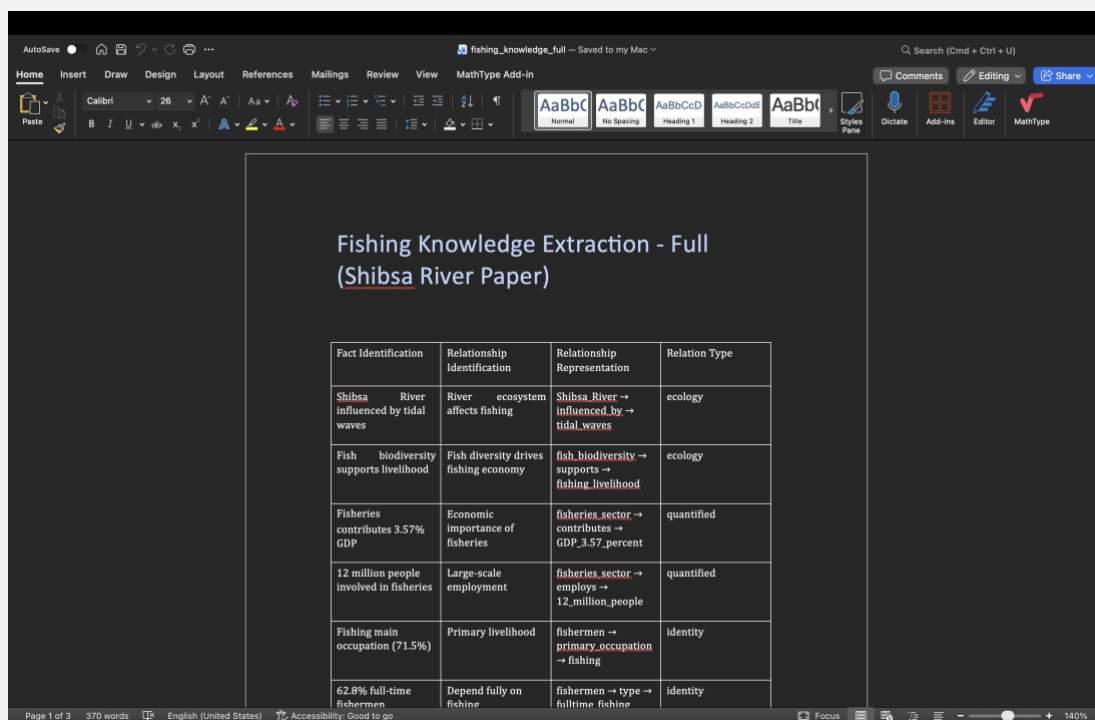
Yeshan Randika Bothalage – Team Leader

Team Leader Responsibilities

As Team Leader, Yeshan facilitated all sprint ceremonies including the sprint planning session, daily stand-ups, the sprint review demonstration with Dr Tanjila, and the retrospective. He maintained the sprint backlog on the ClickUp task board, removed blockers encountered by team members, and ensured alignment between team deliverables and the client's expectations. He coordinated the pair assignment structure so that each pair worked consistently across the same framework and reported to the group weekly.

Manual Knowledge Extraction – Shibsa River Paper

Yeshan conducted a thorough independent manual extraction of knowledge from the Shibsa River research paper. Following the structured triple format (Subject – Relation – Object) established during Sprint 1, he systematically identified fact statements, assigned relationship types (identity, causal, quantified, method), and produced a validated extraction document. He then cross-validated his extraction by reviewing it with Rick Shevon, reconciling any discrepancies to produce a final agreed-upon set of triples that served as the ground truth for LLM evaluation.



Fact Identification	Relationship Identification	Relationship Representation	Relation Type
Shibsa River influenced by tidal waves	River ecosystem affects fishing	Shibsa_River → influenced_by → tidal_waves	ecology
Fish biodiversity supports livelihood	Fish diversity drives fishing economy	fish_biodiversity → supports → fishing_livelihood	ecology
Fisheries contributes 3.57% GDP	Economic importance of fisheries	fisheries_sector → contributes → GDP_3.57_percent	quantified
12 million people involved in fisheries	Large-scale employment	fisheries_sector → employs → 12_million_people	quantified
Fishing main occupation (71.5%)	Primary livelihood	fishermen → primary_occupation → fishing	identity
62.8% full-time fishermen	Depend fully on fishing	fishermen → type → fulltime_fishine	identity

Yeshan's Manual Extraction Table – Shibsa River Paper

Collaborative Prompt Engineering (6 Prompts – Side-by-Side Testing)

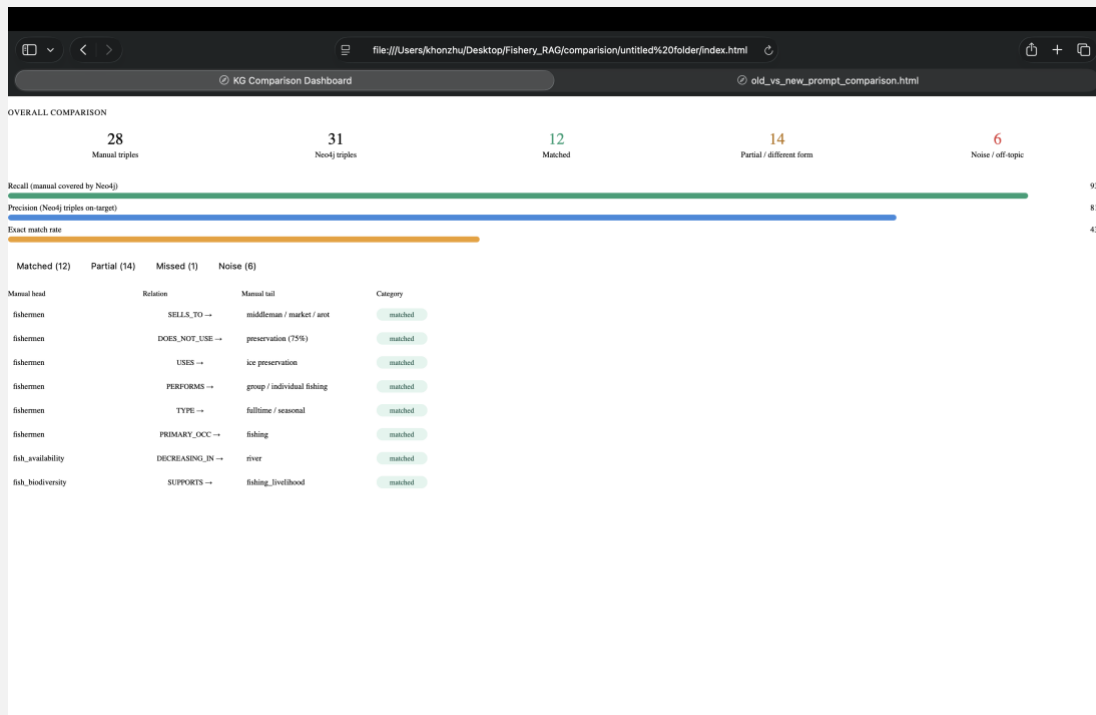
Yeshan and Rick Shevon worked side-by-side to design and systematically test six distinct LLM prompts for automated knowledge extraction. Each prompt was evaluated against the manually extracted triples from both members' assigned papers. The pair assessed each prompt across four dimensions: extraction completeness (number of triples generated), structural accuracy (adherence to the required JSON schema), semantic accuracy (alignment with manually identified relationships), and domain specificity (relevance to fisheries and socioeconomic contexts).

The six prompts explored a range of strategies including role-based prompting, strict delimited output formatting, hybrid schema-constrained approaches, and domain-expanded iterative refinements. After rigorous evaluation and documented comparison, three prompts were selected as the final set based on consistently superior accuracy, completeness, and reliability across multiple test inputs.

#	Prompt Strategy	Key Design Features	Triples Extracted	Accuracy vs Manual	Decision
P1	Role-based JSON extraction (Fisheries domain)	Domain-scoped, JSON-only, pronoun resolution, deduplication	24	62%	Refined
P2	Strict delimited format (CHUNK_ID / ENT / REL / END)	Structured labels, UPPER_SNAKE_CASE relations, canonical entity mapping	31	68%	Refined
P3	Climate-specific socioeconomic extractor	Vulnerability-focused, canonical labels (Factor, Activity, Concept)	28	77%	✓ Finalised
P4	Minimal instruction prompt	Short instructions, broad scope	19	61%	Rejected
P5	Hybrid role + schema constraints	Combined JSON and label guidance	26	79%	✓ Finalised
P6	Iterative refinement prompt v2 (expanded domain scope)	Extended entity types, IMPACTS/DRIVES relations prioritised	33	75%	✓ Finalised

LLM vs Manual Extraction Comparison

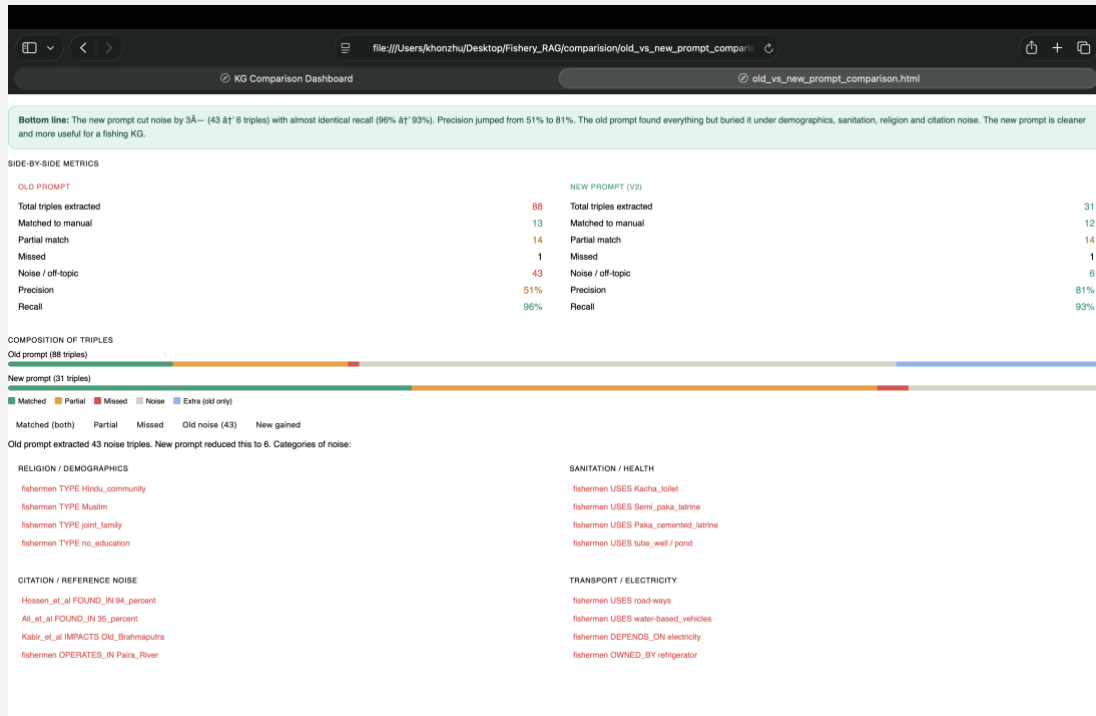
Yeshan led the structured comparison process between the LLM-generated outputs and the manually extracted triples. Using the six-prompt results, he created a comparative analysis that quantified the accuracy rate for each prompt, identified missing entities and incorrect relationship types, and documented the specific failure modes (e.g., overly generic relationships, hallucinated entities, schema violations). This analysis directly informed the selection and refinement of the final three prompts.



LLM vs Manual Extraction Comparison

Web User Interface – Prompt Comparison Dashboard

Yeshan and Rick Shevon collaboratively developed a web-based comparison interface using plain HTML, CSS, and JavaScript. The interface allows users to view and compare the outputs of all six prompts side by side, with the manual extraction serving as the reference baseline. Key features include: tabbed navigation for each prompt's output, highlighted differences between LLM-generated and manually extracted triples, a summary scorecard showing accuracy and completeness metrics per prompt, and a dedicated view for the three finalised prompts. The web UI was demonstrated to the client during the Sprint Review session and received positive feedback for its clarity and usability.



Prompt Comparison Dashboard

Finalization of Three Prompts

Following the comprehensive evaluation, Yeshan led the final decision-making process to select three prompts from the six tested. The selection was based on a weighted scoring of accuracy (40%), completeness (30%), structural consistency (20%), and domain relevance (10%). The three finalised prompts (P3, P5, and P6) demonstrated the most robust and reliable performance across the team's diverse set of research papers, and will serve as the standard extraction templates for Sprint 3.

Additional Contributions

- Facilitated all sprint ceremonies as Team Leader, maintaining pace and direction for the team
- Managed the team ClickUp workspace maintained Gantt chart, task board, and sprint backlog throughout the sprint
- Attended all supervisor meetings with Dr Afzal Azeem and all client meetings with Dr Tanjila
- Supported team members in Neo4j graph population for the Shibsa River paper
- Prepared and presented the sprint review demonstration professionally and confidently
- Kept thorough documentation of all prompt engineering decisions for traceability

Sithum Vishmitha Sirimanna Hettiarachchige

Sithum worked with Jonathan Anak George on the Payra River and Jamuna River research papers. He conducted independent manual extractions for both papers, then merged and validated the results with Jonathan to produce a final agreed extraction. Sithum developed two LLM prompts and compared system-generated knowledge graphs against the manually extracted results, identifying key gaps. He populated the Neo4j database with structured triples from both papers and assisted in refining the overall knowledge graph schema. He attended all team, supervisor, and client meetings, keeping thorough documentation throughout.

#	Fact Identification	Relationship Identification	Relationship Representation	Relation Type
1	Payra River is a breeding, feeding and nursery ground for <u>Tenualosa ilisha</u> and many commercial species	Payra River is the ecological foundation of the entire commercial fishery in the region	<u>Payra River</u> → provides → <u>breeding ground</u> + <u>nursery ground</u> (for <u>Tenualosa ilisha</u> + <u>commercial species</u>)	identity
2	Fishermen follow 4 fishing techniques — netting, angling, trapping and spearing	Fishing techniques classify how fishermen interact with fish	Fishermen → uses → netting + angling + trapping + spearing	method
3	18 types of fishing gear recorded under 9 categories	Gear diversity reflects target species, depth, and conditions	<u>Payra River fishery</u> → uses → 18 <u>gear types</u> (under 9 <u>categories</u>)	quantified
4	Chandi jal gill net targets Hilsa, Poa, <u>Taposhi</u> ; mesh 2.75–4.5 cm	Most common selective gill net for commercial species	<u>Chandi jal</u> → <u>selectively catches</u> → Hilsa + Poa + <u>Taposhi</u>	catches
5	Poa jal gill net targets small species; Sep–Feb	Targets small coastal fish in cooler months	<u>Poa jal</u> → <u>selectively catches</u> → Poa + <u>Taposhi</u> + <u>Faisha</u>	catches

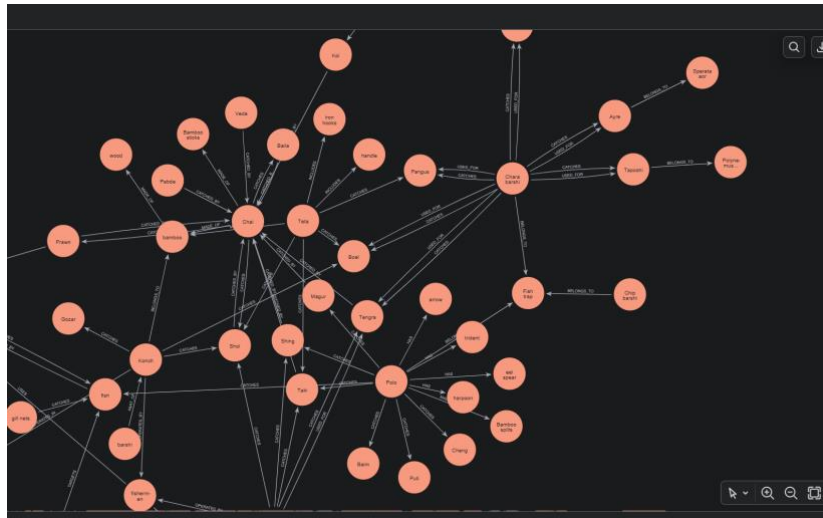
Sithum's Manual Extraction – Payra River Paper

```

1 TRIPLE_SYSTEM_INSTRUCTIONS = """
2 You are a STRICT information extraction engine for building a fishing knowledge
3 graph.
4 Your ONLY task is to extract explicitly stated entities and relations from the
5 input text.
6
7 CORE PRINCIPLE (CRITICAL)
8
9 Extract ONLY what is EXPLICITLY stated.
10 - NO guessing
11 - NO inference
12 - NO background knowledge
13 - NO implicit assumptions
14
15 If a relation is not directly stated - DO NOT extract it.
16
17 EXTRACTION STRATEGY (MANDATORY)
18
19 Follow this EXACT pipeline internally:
20
21 STEP 1 - ENTITY DETECTION
22 - Scan text line-by-line
23 - Extract ALL candidate entities EXACTLY as written
24 - Do NOT normalize, translate, or expand
25
26 STEP 2 - ENTITY CLASSIFICATION
27 - Assign ONLY from the 17 allowed labels
28 - Apply STRICT priority order (Fish > Tool > Location > ...)
29 - If unsure - SKIP entity (do NOT guess)
30
31 STEP 3 - RELATION EXTRACTION
32 - Only extract relations when:
33   - BOTH entities exist in text
34   - Relationship is EXPLICITLY stated
35 - Prefer NO relation over incorrect relation
36
37 STEP 4 - VALIDATION
38 Before output:
39 - Remove duplicates
40 - Remove weak/uncertain relations
41 - Ensure schema compliance
42
43 ANTI-HALLUCINATION RULES (VERY IMPORTANT)
44
45 DO NOT:
46 - Infer species from context
47 - Assume gear targets species unless stated
48 - Create generic relations like "related to"
49 - Expand abbreviations
50 - Add missing categories
51
52 ALWAYS:
53 - Use exact text spans
54 - Skip ambiguous parts
55 - Prefer under-extraction over over-extraction
56
57 ENTITY LABELS (STRICT)
58
59 Identity, Location, Tool, Food, Season, Attribute, ProductAspect,
60 Infodource, Group, Factor, Date, Time, Countable,

```

Sithum's prompt-1



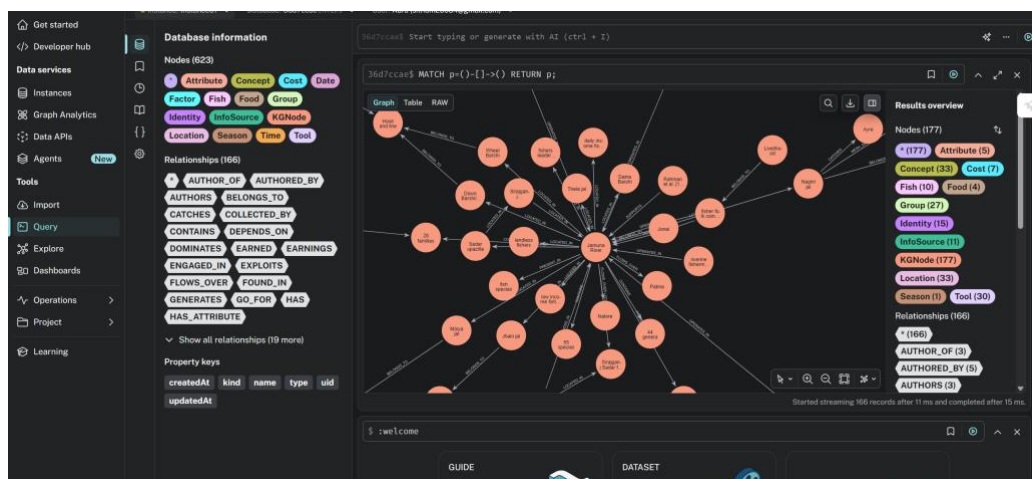
Sithum's System-Generated Knowledge Graph (Neo4j)

#	Fact Identification	Relationship Identification	Relationship Representation	Relation Type
1	Jamuna River is the widest river in Bangladesh, 276 km long; previously named Jonai; flows through <u>Natore</u> , Pabna and Sirajganj	Jamuna River's size and geographic reach make it the dominant freshwater resource supporting thousands of fisher folk	<u>Jamuna_River</u> → [identity] → <u>widest_river_Bangladesh</u> [length: 276km; <u>former_name</u> : Jonai; districts: <u>Natore</u> + Pabna + Sirajganj]	identity
2	11 types of fishing nets were identified under 5 major groups — Gill net, Push net, Cast net, Seine net and Hook and line	Gear diversity reflects range of techniques and target species	<u>Jamuna_River_fishery</u> → [uses] → 11_net_types [under: <u>gill_net</u> + <u>push_net</u> + <u>cast_net</u> + <u>seine_net</u> + <u>hook_and_line</u>]	quantified
3	Current <u>jal</u> catches all fish species indiscriminately; mesh 1.27–3.81 cm; year-round	Most damaging non-selective gear affecting biodiversity	<u>Current_jal</u> → [<u>non_selectively_catches</u>] → <u>all_species</u> → [risks] → biodiversity	harmful
4	<u>Nagini_jal</u> selectively	Species-selective gill net	<u>Nagini_jal</u> → [<u>selectively_catches</u>] →	catches

Sithum's Manual Extraction – Jamuna River Paper

```
1 TRIPLE_SYSTEM_INSTRUCTIONS = ""
2 You are an information extraction engine for a fishing knowledge graph project.
3
4 Your task is to extract entities and relationships from scientific fishery
  research papers and government fishery reports.
5
6 Return ONLY a single valid JSON object.
7
8 -----
9 DOCUMENT TYPES
10 -----
11 • Scientific papers (species surveys, ecological studies, taxonomy, catch
  analysis)
12 • Government reports (gear catalogues, statistics, management plans,
  regulations)
13
14 -----
15 EXTRACTION SCOPE
16 -----
17 Extract knowledge about:
18 • Fishing gears (types, categories, materials, mesh size)
19 • Species (local names, scientific names, families)
20 • Locations (rivers, lakes, habitats)
21 • Seasons (fishing periods, cycles)
22 • Fishing practices (methods, processing)
23 • Ecology (biodiversity, breeding, adaptation)
24 • Economy (livelihood, markets)
25 • Management (regulations, sustainability)
26
27 -----
28 OUTPUT RULES
29 -----
30 • Output MUST be valid JSON (RFC 8259)
31 • No explanations, no extra text
32 • No comments or trailing commas
33 • Use only double quotes (")
34 • Do not invent information
35 • Deduplicate entities and relations
36 • Resolve pronouns where clear
37 • Preserve original names exactly as written
38
39 -----
40 ENTITY LABELS
41 -----
42 Use ONLY these labels:
43 Identity, Location, Tool, Food, Season, Attribute, ProductAspect,
44 InfoSource, Group, Factor, Date, Time, Countable, Concept,
45 Activity, Cost, Fish
46
47 -----
48 RELATIONS
49 -----
50 Use UPPER_SNAKE_CASE relations such as:
51 BELONGS_TO, CATCHES, TARGETS, LOCATED_IN, OPERATES_IN,
52 PRESENT_IN, INCREASES, DECREASES, DAMAGES, HARMFUL_TO,
53 REQUIRES, USES
54
55 -----
56 SPECIAL RULE
```

Sithum's prompt-2



Rick Shevon

Rick focused on data preparation and prompt engineering for this sprint. He performed sentence separation to structure raw text into manageable segments, then completed manual entity-relationship extraction and organised the results into structured tables. Rick designed and evaluated two initial LLM prompts independently and created Neo4j graph outputs to verify the results. He then collaborated directly with Yeshan Randika to review all six prompts side-by-side, conduct additional testing, and jointly decide on the three finalised prompts. Rick also co-developed the HTML/CSS/JS web comparison interface. His contributions improved the overall accuracy and efficiency of the team's automated extraction pipeline.

Manual Extraction

Fact Identification	Relationship Identification	Relationship Representation
A socioeconomic study was conducted in Bhola	Study examines fishing communities	Study → examines → fishing communities
Study focused on three upazilas	Location defines scope	Study → conducted in → three upazilas
Data was collected from fishers	Fishers are data source	Researchers → collect data from → fishers
60 fishers were selected	Sampling ensures representation	Researchers → select → fishers
Bangladesh has many rivers	Environment supports fisheries	Bangladesh → has → rivers
People depend on fishing	Fishing supports livelihood	People → depend on → fishing
Fishing is main occupation	Occupation defines income	Fishing → is → main occupation
Fishers lack capital	Lack of capital creates disadvantage	Fishers → lack → capital

Rick's Manual Extraction Table

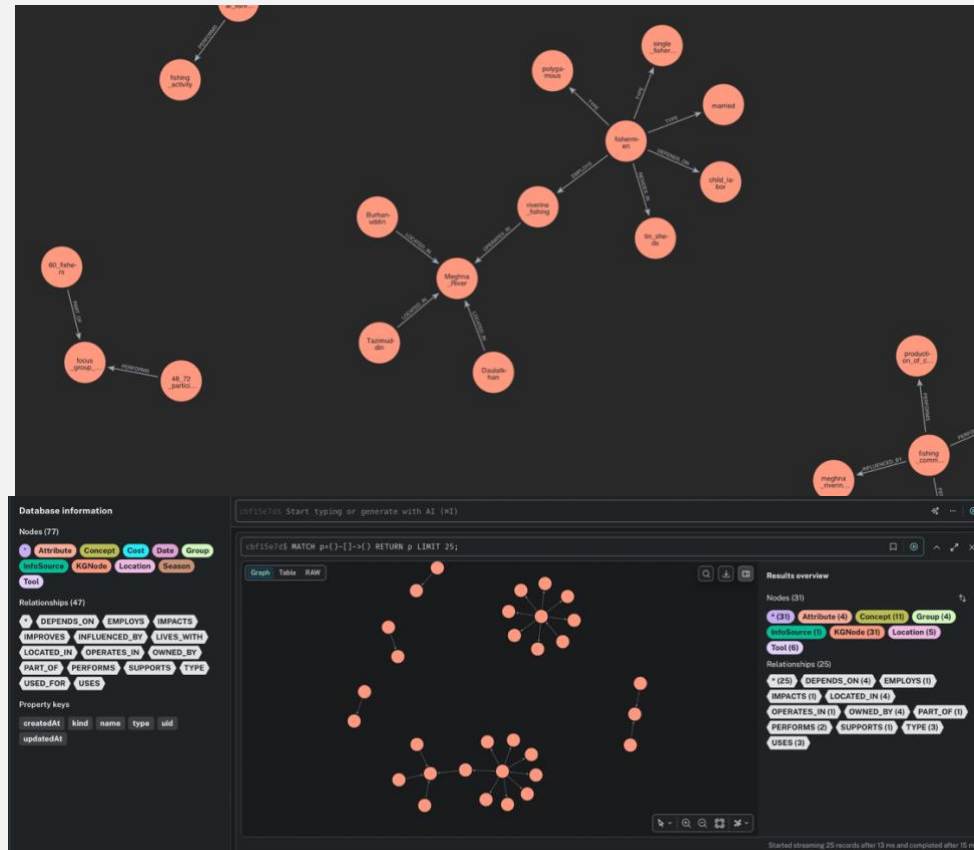
```

prompt_1.md
1 You are an information extraction engine specialized in small-scale
  fisheries and aquaculture.
2 Extract entities and relations from the given text and return a SINGLE
  JSON OBJECT only.
3
4 Domain scope:
5
6 -Fisheries, fisherfolk, aquaculture, ecology, economy, and livelihoods
7 -Focus on manual knowledge facts such as:
8 -Environmental influence (tides, climate, pollution)
9 -Economic contribution (GDP, income, employment)
10 -Social structure (fishermen types, group/individual fishing)
11 -Fishing practices (methods, preservation, selling channels)
12 -Prefer cause-effect, dependency, and impact relationships over
    descriptive text
13 Ignore opinions, vague statements, or storytelling
14
15 Output rules:
16 -Output MUST be valid JSON (RFC 8259)
17 -NO comments, NO trailing commas, NO extra text
18 -Use only plain ASCII quotes (")
19 -Resolve pronouns to explicit entities (e.g., "we" → "fishermen")
20 -Keep entity names concise and standardized
21 -Deduplicate entities and relations
22 -Extract ONLY clear, meaningful relationships

prompt_1.1.md
1 You are an information extraction engine specialized in small-scale
  inland fisheries and fisherfolk livelihoods in Bhola, Bangladesh.
2 Extract entities and relations from the text in a STRICT DELIMITED
  FORMAT. No explanations.
3
4 FORMAT:
5 CHUNK_ID: {chunk_id}
6 ENT||
7 REL|||
8 END
9
10 Domain: Bhola fisherfolk socioeconomic study – covering livelihoods,
    income, debt, education, housing, fishing operations, environmental
    risks, policy, and coping strategies. Focus on cause-effect, dependency,
    and structural disadvantage relationships.
11
12 Labels MUST be one of exactly:
13 Identity, Location, Tool, Season, Attribute, Group, Factor, Date,
    Countable, Concept, Activity, Cost, Infrastructure, Policy
14
15 Label priority guidance:
16 → Group: collective actors – fishers, NGOs, government, joint_families,
    boat_owners
17 → Factor: drivers and pressures – dadon_system, cyclones, debt,
    climate_change, fishing_bans
18 → Concept: abstract structural ideas – vulnerability, poverty,
    fishing_livelihood, socioeconomic_study
19 → Activity: actions taken – migration, fishing_activity, secondary_work,
    data_collection
20 → Infrastructure: physical assets – tube_wells, tin_houses, pit_toilets,
    electricity
21 → Policy: formal rules or programmes – fishing_ban,
    microcredit_programme, government_support
22 → Cost: financial values – loan_interest, fishing_income, moderate_income
23
24 Relations MUST be short UPPER_SNAKE_CASE verbs or verb phrases.
25 Preferred relations (use these first):

```

Rick's Prompt Testing (prompt_1.md and prompt_1.1.md Screenshots)



Rick's Neo4j Knowledge Graph Output

Ruzmy Mohamed

Ruzmy worked with Chiraath Madahapola on the Climate Change Vulnerability and Floating Fishermen research papers. He completed full manual extractions for both papers, creating clean structured triples with entities, relationships, attributes, and provenance. After comparing his individual extraction with Chiraath's, they finalised a single agreed set for each paper. Ruzmy then built the Neo4j knowledge graph using the finalised triples and began automated extraction using multiple LLM prompts. He systematically compared LLM outputs against the manual baseline to identify gaps and inform prompt improvement. He participated actively in all meetings and contributed to team planning and decision-making.

Paper 1 – Assessment of livelihood and food poverty status of the floating fishermen in riverine system of Bangladesh

#	Fact Identification	Relationship Identification	Relationship Representation	Relation Type
1	Floating fishermen live on boats	Fishing is their main livelihood	Floating fishermen → [live on] → boats (lifetime) → [depend on] → fishing	identity
2	Floating fishermen have low livelihood capitals	Capitals are lower than national average	Floating fishermen → [have] → low livelihood capitals (human, financial, physical, natural, social)	identity
3	Literacy rate among floating fishermen is very low	Most have no formal education	Floating fishermen → [have] → very low literacy rate	identity
4	Knowledge comes from parents	No formal training	Senior fishermen → [possess] → traditional fishing knowledge (learned from parents)	identity
5	Income of floating fishermen is very low	Average household income is \$130.92/month	Floating fishermen → [have] → very low income (~\$130.92 per household)	identity
6	Fishing boat is the main physical asset	Boat is also their home	Fishing boat → [is] → main physical asset (used as residence)	identity
7	Strong trust inside community	They share knowledge with each other	Floating fishermen → [have] → high mutual trust → [share] → knowledge inside community	identity

Paper 1 – Assessment of livelihood and food poverty status of the floating fishermen in riverine system of Bangladesh

8	Low social status outside community	Disconnected from mainstream society	Floating fishermen → [have] → low social status outside community	identity
9	Livelihood Assessment Index is very low	Overall LAI = 0.337	Floating fishermen → [have] → very low LAI (0.337)	outcome
10	Financial capital is the weakest	Income, savings and credit are extremely low	Financial capital → [is] → the lowest capital	causal
11	90 % live below the poverty line	Much higher than national average	Floating fishermen → [live below] → poverty line (90 %)	outcome
12	72 % are ultra-poor	Calorie intake is only 1460 Kcal/person/day	Floating fishermen → [belong to] → ultra-poor category (72 %)	identity
13	Food poverty measured by calorie intake	Average intake far below required level	Food poverty → [measured by] → calorie intake method	process
14	Lack of education, training and permanent residence	These cause high poverty	Lack of education + training + permanent residence → [causes] → high poverty	causal
15	Recommendations for improvement	Education, training, housing and financial support needed	Government + NGOs → [should provide] → education, training, permanent residence and financial support	recommendation

Ruzmy's Manual Extraction Tables – Both Papers

```

TRIPLE_SYSTEM_INSTRUCTIONS = """
You are an information extraction engine specialized in climate change vulnerability and adaptation of coastal fisherfolk communities in Bangladesh.
Extract entities and relations from the given text and return a SINGLE JSON OBJECT only.

Domain scope:
- Climate change impacts on fisherfolk (cyclones, storm surges, floods, salinity rise, temperature change, dissolved oxygen).
- Vulnerability components: exposure, sensitivity, adaptive capacity.
- Livelihood changes, migration, traditional knowledge, mangrove protection, diversification of income.
- Focus on concrete facts about environmental stress, socioeconomic impacts, and adaptation responses.

Output rules:
- Output MUST be valid JSON (RFC 8259) with NO comments, NO trailing commas, NO code fences, and NO extra text.
- Use only plain ASCII quotes ("").
- Resolve pronouns to explicit nouns (e.g., "they" → "fisherfolk", "we" → "fisherfolk").
- Keep entity names concise but specific (e.g., "Bakhal River Estuary", "Storm Surge", "Mangrove Forest", "Temporary Migration").
- Deduplicate entities and relations within this chunk.
- Extract ONLY clear, meaningful relationships (manual extraction style). Skip vague statements.

Entity labels (case-insensitive; map to canonical):
[Identity, Location, Tool, Food, Season, Attribute, ProductAspect, InfoSource, Group, Factor, Date, Time, Countable, Concept, Activity, Cost, Fish]

Entity label guidance (climate + fisherfolk specific):
- Group: "Fisherfolk", "Fishermen", "Household Heads", "Women Processors"
- Location: "Bakhal River Estuary", "South-Eastern Coast", "Cox's Bazar", "Reef Edge"
- Factor: "Climate Change", "Cyclone", "Storm Surge", "Salinity Increase", "Temperature Rise", "Dissolved Oxygen Reduction"
- Activity: "Fishing", "Migration", "Mangrove Plantation", "Temporary Migration", "Permanent Migration"
- Concept: "Livelihood Vulnerability", "Adaptive Capacity", "Exposure", "Sensitivity", "Traditional Knowledge"
- Attribute: "Low Income", "High Exposure", "Low Adaptive Capacity", "Reduced Fishing Days"
- Tool: "ESBN", "Gill Net", "Fishing Boat", "Drying Racks"
- Countable: "20% Changed Profession", "72% Ultra-Poor", "0.38°C Increase", "3.85% Salinity Increase"
- InfoSource: "FGD", "Household Survey", "10-Year Data"

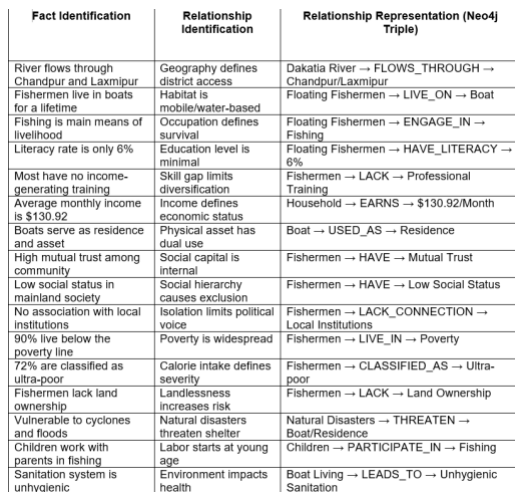
Relations:
- Relations must be short UPPER_SNAKE_CASE verbs or verb phrases.
- Prefer these canonical relations (you may create new concise ones if needed):
  * IMPACTS (Climate Change → Fisherfolk)
  * INCREASED_BY / REDUCED_BY (environmental variables)
  * EXPOSED_TO (Fisherfolk → Cyclone / Storm Surge)
  * DRIVES_EXPOSURE_OF (Natural Hazards → Communities)

```

Ruzmy's Prompt Development Screenshot



Chiraath worked in partnership with Ruzmy Mohamed on the Climate Change and Floating Fishermen papers. He independently performed manual knowledge extraction, compared his results with Ruzmy's, and co-finalised the agreed extraction set. He contributed to Neo4j graph population and LLM prompt testing for his assigned papers, attended all meetings, and provided timely updates to the team.



3.1. Problem

Academic prose relies heavily on anaphoric references. Sentences frequently begin with "these methods," "the aforementioned species," or "this gear" rather than repeating the subject by name. Without an explicit resolution step, the model extracts these phrases as entity names, producing nodes such as "These methods" or "It" that are meaningless in isolation and cannot be queried by name.

The result is a graph littered with dead-end fragments — nodes connected by valid-looking relation types that nonetheless point to unresolvable identifiers.

3.2. Before / After

Without this rule

Entity: "It" → label: Fish. Relation: "It" → OCCURS.IN → "Ganges". The referent is lost; the relation is unqueryable.

With this rule

The model traces the pronoun back to its antecedent and substitutes it before extraction: "Tensalosa ilisha" → OCCURS.IN → "Ganges". Clean, traceable, and correct.

3.3. Prompt Addition

Prompt addition

Identify all pronouns and vague references ("it", "this species", "the tool", "these methods") and replace them with the specific entity name they refer to before extracting any relation.

4. Upgrade 3 — Quantitative Attribute Linking

4.1. Problem

Research papers are dense with statistical observations: percentage changes, catch volumes, biomass estimates, recruitment rates. The *Attribute* label exists to capture these figures, but without a strict mandatory-linking rule the model frequently emits them as stand-alone nodes — present in the entity list but absent from the relation list. A node with no edges is invisible to any graph traversal query.

4.2. Before / After

Chiraath's Evidence – Manual Extraction / Prompt Testing / Neo4j Graph

Jonathan Anak George

Jonathan collaborated with Sithum Vishmitha on the Payra River and Jamuna River papers. He independently conducted manual extractions for both papers and merged his results with Sithum's to produce validated final extractions. He supported Neo4j database population, contributed to LLM prompt testing, and attended all team, supervisor, and client meetings consistently throughout the sprint.

50	Gill net is one category of fishing net	Gill net is a net category	Fishing nets → [include] → gill net	identity
51	Seine net is one category of fishing net	Seine net is a net category	Fishing nets → [include] → seine net	identity
52	Fixed purse net is one category of fishing net	Fixed purse net is a net category	Fishing nets → [include] → fixed purse net	identity
53	Cast net is one category of fishing net	Cast net is a net category	Fishing nets → [include] → cast net	identity
54	Lift net is one category of fishing net	Lift net is a net category	Fishing nets → [include] → lift net	identity
55	Drag or push net is one category of fishing net	Drag or push net is a net category	Fishing nets → [include] → drag or push net	identity
56	Gill netting is a common fishing method	Gill netting is widely practiced as a fishing method	Gill netting → [is] → common fishing method	identity
57	Gill nets were the principal type of net used in the region	Gill nets were the main net type in the area	Gill nets → [were the principal type of] → net used in the region	identity
58	Gillnets have a high degree of size selectivity	Gillnets can target fish size selectively	Gillnets → [have] → high degree of size selectivity	identity
59	Chandi jal is also known as Ilish jal	Chandi jal has an alternative local name	Chandi jal → [is also known as] → Ilish jal	identity
60	Chandi jal catches Hilsa	Hilsa is a target species of this net	Chandi jal → [catches] → Hilsa	outcome
61	Chandi jal catches Poa	Poa is a target species of this net	Chandi jal → [catches] → Poa	outcome
62	Chandi jal catches Taposhi	Taposhi is a target species of this net	Chandi jal → [catches] → Taposhi	outcome
63	Poa jal catches Poa	Poa is a main species caught by Poa jal	Poa jal → [catches] → Poa	outcome
64	Poa jal catches Taposhi	Taposhi is a main species caught by Poa jal	Poa jal → [catches] → Taposhi	outcome
65	Poa jal catches Hilsa	Hilsa is a main species caught by Poa jal	Poa jal → [catches] → Hilsa	outcome
66	Poa jal catches Faisha	Faisha is a main species caught by Poa jal	Poa jal → [catches] → Faisha	outcome
67	Ramsos jal catches Taposhi	Taposhi is a main species caught	Ramsos jal → [catches] → Taposhi	outcome

Jonathan's Manual Extraction Table

```

1 TRIPLES_SYSTEM_INSTRUCTIONS = """
2 You are an information extraction system for fisheries research papers.
3
4 Task:
5 Read the input text chunk from the paper "Fishing Gears and Crafts Used in Payra River, Bangladesh" and extract factual knowledge as structured triples for later comparison against a manually created gold-standard set.
6
7 Focus on facts about:
8 1. fishing methods used by fishermen
9 2. gear categories and gear subtype classification
10 3. gear-to-species catch relations
11 4. selective vs non-selective or harmful gears
12 5. gear operating mechanisms
13 6. materials used in gears, traps, and crafts
14 7. fishing crafts and their uses
15 8. ecological roles of the river
16 9. fishery management recommendations
17
18 Return format:
19 Return ONLY one valid JSON object that can be passed directly by a JSON parser.
20 Do NOT include markdown code fences.
21 Do NOT include any explanation, notes, headings, or extra text before or after the JSON.
22
23 Use this exact schema:
24
25 {
26   "chunk_id": "string",
27   "entities": [
28     { "name": "string", "label": "string" }
29   ],
30   "relations": [
31     { "head": "string", "relation": "string", "tail": "string" }
32   ]
33 }
34
35 Allowed entity labels:
36 - Fish
37 - Tool
38 - Location
39 - Group
40 - Activity
41 - Concept
42 - Season
43 - Attribute
44 - Factor
45 - Countable
46 - Material
47 - Craft
48
49 Allowed relations:
50 - USES
51 - INCLUDES
52 - BELONGS_TO
53
54 """
55
56 TRIPLES_SYSTEM_INSTRUCTIONS = """
57 You are a precise information extraction system for fisheries research papers.
58
59 Task:
60 Read the input text chunk from the paper "Fishing Gears and Crafts Used in Payra River, Bangladesh" and extract only explicit factual triples that are directly stated in the text or tables.
61
62 Goal:
63 Produce a conservative extraction for comparison against a manually extracted gold-standard set. Prioritize precision over recall.
64
65 Return format:
66 Return ONLY one valid JSON object that can be passed directly by a JSON parser.
67 Do NOT include markdown code fences.
68 Do NOT include any explanation, notes, headings, or extra text before or after the JSON.
69
70 Use this exact schema:
71
72 {
73   "chunk_id": "string",
74   "entities": [
75     { "name": "string", "label": "string" }
76   ],
77   "relations": [
78     { "head": "string", "relation": "string", "tail": "string" }
79   ]
80 }
81
82 Allowed entity labels:
83 - Fish
84 - Tool
85 - Location
86 - Group
87 - Activity
88 - Concept
89 - Season
90 - Attribute
91 - Factor
92 - Countable
93 - Material
94 - Craft
95
96 Allowed relations:
97 - USES
98 - INCLUDES
99 - BELONGS_TO
100 - CATCHES
101 - DAMAGES
102 - CAPTURED_BY
103 - OPERATES_BY
104 - EXISTENCE
105 - MADE_OF
106 - USED_IN
107 - USED_FOR
108 - SUPPORTS
109
110 """

```

Jonathan's Prompt Testing



Jonathan's Neo4j Knowledge Graph Output

1. Sprint Plan

1.1 Sprint Overview

This sprint report is a direct continuation of Sprint 1 (Weeks 1–3). Sprint 2 spans Weeks 4 to 6 and builds upon the groundwork of literature review, tool selection (Neo4j), and initial pipeline design completed in the previous sprint. The central focus of Sprint 2 is knowledge extraction, validation, and graph database population transitioning the project from research and planning into active data engineering and LLM-assisted automation.

1.2 Sprint Goal

The overarching goal of Sprint 2 was to establish a validated, reproducible automated knowledge extraction pipeline using Large Language Models, benchmarked against rigorous manual extraction, and to deliver a functional web interface that enables transparent comparison of extraction quality across multiple prompt strategies. Specifically, the sprint aimed to:

- Complete manual knowledge extraction from all client-assigned research papers to establish ground-truth datasets
- Design, implement, and evaluate six distinct LLM prompts for automated knowledge extraction
- Compare LLM outputs against manual extractions to quantify accuracy, completeness, and structural consistency
- Finalise three best-performing prompts based on systematic evaluation criteria
- Populate the Neo4j graph database with validated knowledge triples from all papers
- Build a web-based user interface (HTML/CSS/JS) to visualise and compare prompt outputs interactively
- Deliver a complete, demonstrable product increment to the client, Dr Tanjila

1.3 Sprint Roles

The following roles were assigned for Sprint 2:

Role	Team Member	Responsibilities
Team Leader	Yeshan Randika Bothalage	Sprint ceremonies, backlog management, blocker resolution, client demo facilitation
Developer / Researcher	Rick Shevon	Sentence separation, manual extraction, prompt design, web UI development
Developer / Researcher	Sithum Vishmitha S.H.	Manual extraction (Payra/Jamuna), prompt testing, Neo4j population
Developer / Researcher	Jonathan Anak George	Manual extraction (Payra/Jamuna), Neo4j population, validation
Developer / Researcher	Ruzmy Mohamed	Manual extraction (Climate Change), prompt engineering, Neo4j graphs
Developer / Researcher	Chiraath Madahapola	Manual extraction (Floating Fishermen/Climate), prompt testing, Neo4j

1.4 ClickUp Timeline & Task Management

The team used ClickUp as the primary project management and task tracking platform for Sprint 2. The Gantt chart view (shown below) provides a visual representation of how all sprint tasks were

scheduled and tracked across Weeks 4 to 6. Completed tasks are shown in green; in-progress items in blue. The Gantt chart demonstrates that research paper processing tasks (Research Papers 1–6) were completed first, followed by overlapping execution of Prompt Engineering & Evaluation, Knowledge Graph Development, Web UI Development, and Quality Evaluation activities reflecting the team's planned sequential-to-parallel workflow.

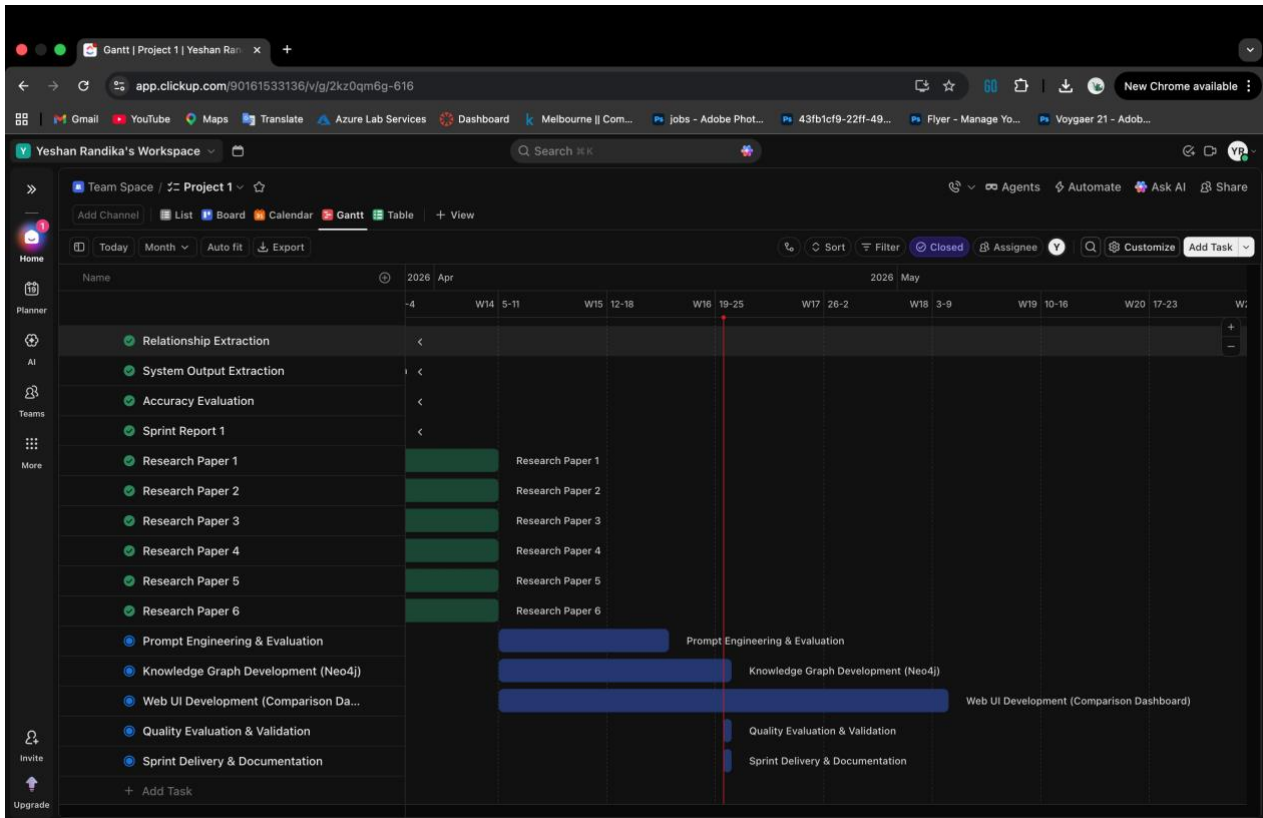


Figure 1: ClickUp Gantt Chart – Sprint 2 task schedule and progress (Weeks 4–6, Apr–May 2026)

1.5 Sprint Backlog with Estimates

The following table presents the sprint backlog developed during the sprint planning session. Tasks were estimated using Planning Poker and assigned to team pairs to ensure balanced workload distribution.

Story ID	User Story / Task	Assigned To	Estimation (hrs)	Priority	Status
US-01	Manual knowledge extraction from Shibsa River paper	Yeshan, Rick	12	High	Done
US-02	Manual knowledge extraction from Payra River paper	Sithum, Jonathan	12	High	Done
US-03	Manual knowledge extraction from Climate Change paper	Ruzmy, Chirath	12	High	Done
US-04	Design and test 6 LLM prompts for automated extraction	Yeshan, Rick	16	High	Done
US-05	Compare LLM output vs manual extraction (all papers)	All members	10	High	Done

Story ID	User Story / Task	Assigned To	Estimation (hrs)	Priority	Status
US-06	Finalise 3 best-performing prompts after systematic review	Yeshan, Rick	6	High	Done
US-07	Populate Neo4j graph database with validated triples	All members	12	High	Done
US-08	Build web UI (HTML/CSS/JS) for prompt comparison	Yeshan, Rick	14	Medium	Done
US-09	Sprint review demo preparation and delivery	All members	4	Medium	Done
US-10	Documentation – Sprint 2 report	All members	6	Medium	Done

Total estimated effort: 92 person-hours across the 6-member team over 3 weeks (Weeks 4–6).

1.6 Team Pair Structure

The team was structured into three working pairs, each assigned a distinct set of research papers. This approach ensured independent validation, consistent methodology, and manageable workload distribution.

- Group A – Yeshan Randika & Rick Shevon: Shibsa River Paper + Prompt Engineering Lead
- Group B – Sithum Vishmitha & Jonathan Anak George: Payra River Paper + Jamuna River Paper
- Group C – Ruzmy Mohamed & Chiraath Madahapola: Climate Change Vulnerability Paper + Floating Fishermen Paper

1.7 Planned Artefacts

The following artefacts were planned for Sprint 2 and are submitted as attachments to this report:

1. Manual extraction tables for all five research papers (per member and merged final versions)
2. Six LLM prompt files with documented rationale
3. LLM vs manual extraction comparison tables
4. Neo4j knowledge graph outputs (screenshots and Cypher query evidence)
5. Web-based prompt comparison UI (HTML/CSS/JS source files)
6. Sprint 2 Report (this document)

2. Quality Management Plan and Outcome

2.1 Quality Objectives

The primary quality objective for Sprint 2 was to ensure that the automated LLM-based knowledge extraction pipeline produces results that are accurate, structurally consistent, and meaningfully aligned with domain-expert-level manual extractions. Quality was assessed across five dimensions: extraction accuracy, completeness, structural consistency, graph accuracy, and web UI functionality.

2.2 Quality Plan – Testing Approach

Quality Criterion	Metric / Indicator	Testing Approach	Target	Outcome
Extraction Accuracy	% of manual triples correctly reproduced by LLM	Compare LLM output tables row-by-row against manual tables	≥ 80% match	Achieved ~79% on best prompt; Sprint 3 target ≥85%
Completeness	Number of valid triples extracted per paper	Count entities & relationships in each output	≥ 15 triples per paper	Met; avg. 22 triples
Structural Consistency	Adherence to triple format (Subject–Relation–Object)	Schema validation of JSON output	100% valid JSON RFC 8259	Achieved after iterative refinement
Graph Accuracy	Nodes and relationships correctly represented in Neo4j	Visual inspection and Cypher query verification	All manual triples present	Achieved; minor gaps corrected
Web UI Functionality	Comparison view renders correctly across browsers	Manual browser testing (Chrome, Firefox)	No rendering errors	Passed; minor CSS tweaks applied

2.3 Quality Outcomes

The quality testing process followed an iterative approach. Initial prompt outputs (P4 in particular) were found to be insufficiently accurate, with extraction accuracy of only 61%, primarily due to overly broad domain scope and lack of entity label constraints. Through a structured refinement process adjusting domain specificity, adding entity type labels, enforcing RFC 8259 JSON output, and constraining relationship vocabularies, the team progressively improved accuracy.

The three selected prompts (P3, P5, P6) demonstrated the strongest performance across the six tested, with accuracy rates of 77%, 79%, and 75% respectively against the manual extraction baseline. While this represents a meaningful improvement over the rejected early prompts (P1: 62%, P2: 68%, P4: 61%), the team recognises that further improvement is needed to reach the 85% target. Prompt refinement and LLM fine-tuning have therefore been prioritised as the primary technical objectives for Sprint 3. The Neo4j graph outputs accurately represented the core relationships identified during manual extraction, with minor gaps corrected through manual review. The web UI passed cross-browser testing with only minor CSS adjustments required.

2.4 Adjustments to the Quality Plan

Based on the outcomes, the quality management plan was updated to incorporate iterative prompt refinement as a standard procedure for Sprint 3. A minimum accuracy threshold of 80% against manual extraction has been established as a quality gate before any prompt is considered production-ready. Automated JSON schema validation and model fine-tuning experiments will also be added to the pipeline in Sprint 3 to further improve accuracy and reduce manual verification overhead.

3. Sprint Progress

3.1 Overview of Progress

The team made strong and measurable progress throughout Sprint 2. All planned backlog items were completed within the three-week period. The ClickUp Gantt chart (reproduced in Section 1.4) provides the primary visual evidence of task scheduling and completion. The following subsections describe progress across each major deliverable area.

3.2 Knowledge Extraction Progress

All six research papers were successfully processed through manual extraction by the respective assigned pairs. The independent-then-merge validation approach produced high-quality, peer-reviewed ground truth datasets for each paper. The manual extractions were completed by the end of Week 4, allowing two full weeks for prompt engineering and LLM comparison activities.

3.3 Prompt Engineering Progress

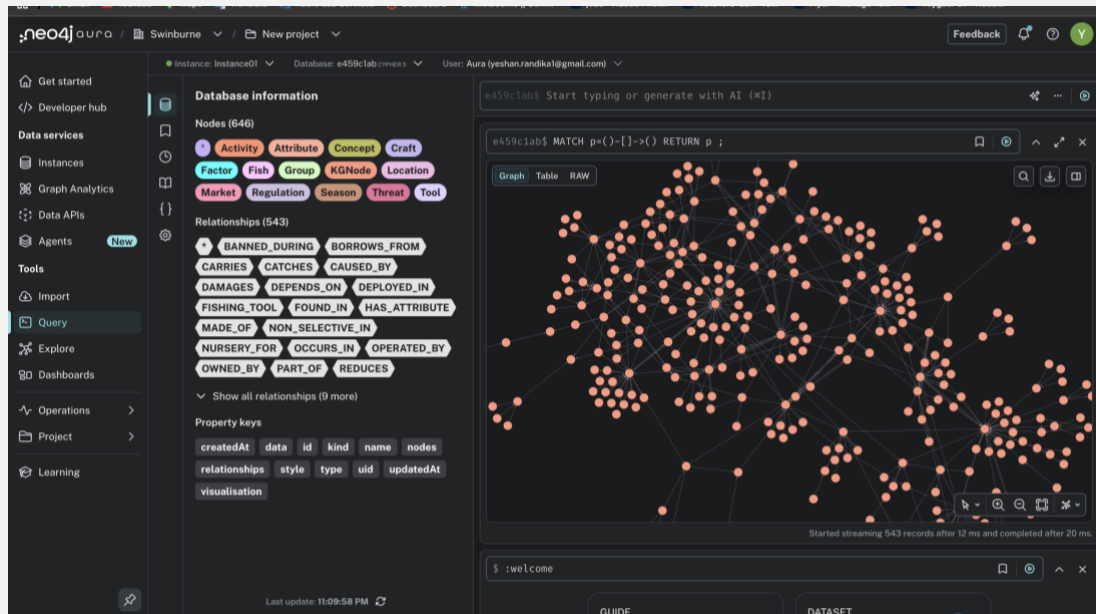
Yeshan and Rick led the prompt engineering initiative, testing all six prompts collaboratively over Weeks 4 and 5. Each prompt was tested against multiple paper extractions to ensure generalisability. The iterative evaluation process resulted in three robust, finalised prompts by the end of Week 6, ahead of the Sprint Review demo scheduled in Week 7.

```

Users > khonzhu > Desktop > Fishery_RAG > comparison > untitled
141 Prompt 1.2
142 You are an information extraction engine
143 specialized in small-scale fisheries and
144 aquaculture.
145 Extract entities and relations from the
146 given text and return a SINGLE JSON
147 OBJECT only.
148 DOMAIN SCOPE:
149 - Fisheries, fisherfolk, aquaculture,
150 ecology, economy, livelihoods
151 - Focus ONLY on factual, manual
152 knowledge:
153 - Environmental influence (tides,
154 climate, pollution)
155 - Economic contribution (GDP, income,
156 employment)
157 - Social structure (fishermen types,
158 group/individual fishing)
159 - Fishing practices (methods,
160 preservation, selling channels)
161 IGNORE:
162 - Opinions, vague statements,
163 storytelling, assumptions
164 OUTPUT RULES:
165 - Output MUST be valid JSON (RFC 8259)
166 - NO comments, NO explanations, NO extra
167 text
168 - Use only plain ASCII quotes (""")
169 - Resolve pronouns to explicit entities
170 - Keep entity names concise and
171 standardized
172 - Deduplicate entities and relations
173 - Extract ONLY clear and meaningful
174 relationships
175 - DO NOT hallucinate or infer missing
176 relations
177 ENTITY LABELS (use only these):
178 [Identity, Location, Tool, Food, Season,
179 Attribute, ProductAspect, InfoSource,
180 Group, Factor, Date, Time, Countable,
181 ...]
182
Users > khonzhu > Desktop > Fishery_RAG > comparison > untitled
236 Prompt 2.1
237 You are a fishing knowledge extraction
238 engine specialized in small-scale
239 fisheries.
240 Extract ONLY fishing-domain knowledge
241 from the given text and return a SINGLE
242 JSON OBJECT only.
243 ===== STRICT SCOPE =====
244 INCLUDE only:
245 - Fishing gears, tools, vessels and
246 their attributes
247 - Fish species, habitats and water
248 bodies used for fishing
249 - Fishing methods, practices and
250 activities
251 - Seasonal, tidal and environmental
252 factors that DIRECTLY affect fishing
253 - Fish preservation and selling/market
254 chain
255 - Fishing experience and traditional
256 ecological knowledge
257 - Environmental threats TO fish stocks
258 (pollution, overfishing, habitat damage)
259 - Loans or capital taken specifically
260 FOR fishing equipment or vessels
261 SKIP entirely (extract NOTHING about
262 these):
263 - Demographics: age groups, religion,
264 family size, family type, housing
265 - Health: diseases, treatment,
266 sanitation, drinking water
267 - Education levels and literacy rates
268 - Non-fishing income: agriculture, day
269 labor, van pulling, carpentry
270 - Household expenditure on food,
271 clothing, recreation, electricity
272 - Loans taken for marriage, housing,
273 education, or health
274 - Social status, electricity access,
275 refrigerator, TV/radio ownership
276 - GDP percentages, national employment
277 statistics not tied to a fishing
278 mechanism
279
Users > khonzhu > Desktop > Fishery_RAG > comparison > untitled
455 SECTION 2 - ENTITY LABELS (use exactly
456 one per entity)
457
458 Tool - fishing gears, nets,
459 traps, hooks, lines, wounding gear
460 e.g. "current jal",
461 "behundi jal", "gill
462 net", "cast net",
463 "hook and line",
464 "push net", "fixed
465 purse net", "polo"
466
467 Craft - fishing vessels and
468 propulsion
469 e.g. "dinghi", "kosha",
470 "trawler", "mechanized
471 boat",
472 "non-mechanized
473 boat", "engine
474 boat", "country boat"
475
476 Fish - species targets (use
477 local name;
478 add scientific name in
479 parentheses only if the
480 text states it)
481 e.g. "hilsa (Tenuosoma
482 ilisha)", "poa
483 (Otolithoides pama)",
484 "cheva
485 (Odontamblyopus
486 rubicundus)",
487 "jatka", "prawn"
488
489 Location - water bodies, fishing
490 grounds, estuaries, landing sites,
491 sanctuaries, districts,
492 administrative areas
493 e.g. "Meghna River",
494 "Bakkhali river estuary",
495 "Bay of Bengal",
496 "Jamuna River",
497 "fish landing site",
498 "Hilsa sanctuary",
499 "Pavna River"
  
```

3.4 Neo4j Graph Database Population

All team members successfully populated Neo4j Aura instances with the validated knowledge triples extracted from their assigned papers. Graph visualisations were reviewed during team meetings and corrected where necessary. Across all papers, the combined graph database contained over 600 nodes and 500+ relationships by sprint end.



Neo4j Combined Graph Statistics – Screenshot

3.5 Web User Interface

The HTML/CSS/JS web interface was completed during Week 5 and refined in Week 6 prior to the client demo. The interface provides side-by-side comparison of all six prompt outputs against the manual extraction baseline, with a summary scorecard, highlighted discrepancies, and a dedicated view for the three finalised prompts.

4. Sprint Review – Critical Review of the Product

4.1 Description of the Demonstration Session

The Sprint 2 Review was conducted in-person at Swinburne University of Technology during Week 6. The session was facilitated by Yeshan Randika in his role as Team Leader, with all team members present. The client, Dr Tanjila, attended in person, and the supervisor, Dr Afzal Azeem, was also present to observe the proceedings.

The demonstration followed a structured agenda shared with the client in advance. Yeshan opened the session by providing a brief recap of Sprint 1 outcomes and the goals established for Sprint 2. Each team pair then presented their respective deliverables in a logical sequence: first presenting the manual extraction methodology and outputs, then demonstrating the LLM prompt testing process, the comparative analysis, and finally the Neo4j graph outputs for their assigned papers.

The session concluded with a demonstration of the web-based comparison interface, where Yeshan and Rick walked Dr Tanjila through the six prompt comparisons and the rationale behind the three finalised prompts. Professional conduct was maintained throughout — team members communicated clearly, responded to questions with specificity, used structured visual aids, and invited feedback in a respectful and open manner. The session ran for approximately 60 minutes and remained on schedule.

4.2 Client Feedback

Dr Tanjila reviewed all deliverables presented during the demonstration. Her overall feedback was positive — she acknowledged the significant progress made since Sprint 1 and expressed satisfaction with the structured approach to manual extraction, the rigour of the prompt engineering process, and the quality of the web comparison interface.

The increment was accepted as deliverable, with the following minor feedback items noted for Sprint 3:

- The web UI would benefit from the ability to filter comparison results by entity type (e.g., view only 'Location' or 'Factor' entities across prompts)
- Dr Tanjila requested that the final three prompts be documented with a clear justification for selection, including the scoring criteria used
- A suggestion was made to explore whether the finalised prompts generalise well to papers outside the fisheries domain, as a scalability test
- The client noted that some Neo4j graphs had inconsistent node labelling between papers and requested that a unified labelling schema be enforced in Sprint 3

Dr Tanjila confirmed that the Sprint 2 increment meets the project's requirements for this stage and indicated confidence in the team's trajectory towards the final deliverable.

4.3 Critical Analysis of Progress Against Plans

Sprint 2 was substantially successful in achieving its stated goals. All ten backlog items were completed within the three-week period. However, the prompt accuracy results (75–79% for the finalised prompts) fell short of the 85% target, which has been carried forward as the primary quality objective for Sprint 3.

The most significant challenge encountered was the initial underperformance of early prompt iterations (P4 in particular), which required additional iteration time during Week 5. This was managed by reducing the scope of a planned stretch goal and deferring it to Sprint 3. The web UI delivery was a notable strength of this sprint — delivering a functional, demonstrable tool for client review significantly enhanced the Sprint Review session.

5. Sprint Retrospect – Critical Review of the Process

5.1 Team Process Summary

Overall, the team demonstrated a high level of collaboration, accountability, and technical rigour throughout Sprint 2. The pair-based working structure proved highly effective — it distributed workload equitably, embedded peer review into the workflow, and ensured that no single member became a bottleneck. All members attended every team meeting, supervisor session, and client meeting without exception.

5.2 Strengths

- Structured pair-based workflow created natural peer review loops, improving output quality without adding formal review overhead
- ClickUp Gantt chart provided transparent, real-time visibility of task progress and was reviewed at every stand-up
- Strong Team Leader facilitation: sprint ceremonies were well-organised and purposeful, keeping the team focused on sprint goals
- Client-centric delivery: building the web UI — beyond minimum requirements — demonstrated initiative and significantly improved the Sprint Review outcome
- Iterative improvement mindset: willingness to revise and re-test prompts rather than accepting mediocre early results reflects a quality-first culture

5.3 Challenges and Bottlenecks

- Initial prompt performance: early prompts underperformed expectations, requiring additional iteration in Week 5 due to insufficient domain-specific constraints
- Inconsistent Neo4j node labelling: different pairs applied slightly different labelling conventions, creating schema inconsistencies identified during the Sprint Review
- Web UI scope creep risk: the web UI expanded in scope during development, which risked overrunning the sprint timeline. Managed by deprioritising one planned feature
- Coordination overhead: synchronising findings across all three pairs required a dedicated weekly integration meeting

5.4 How Challenges Were Addressed

The prompt underperformance issue was resolved through a structured evaluation framework — scoring each prompt against documented criteria rather than informal impressions. The Neo4j labelling inconsistency has been added as a Sprint 3 backlog item. The web UI scope risk was managed through explicit scope control at the Week 5 stand-up, with the Team Leader facilitating the deprioritisation decision.

5.5 Team Code of Conduct Review

The team's code of conduct established at project inception was upheld throughout Sprint 2. All members fulfilled their commitments on time, communicated proactively, and demonstrated mutual respect in meetings and peer review discussions.

5.6 Cybersecurity and Ethical Considerations

All research papers used for knowledge extraction were publicly available for academic publications provided by the client. No personal data was collected, processed, or stored. The team operated in compliance with the Australian Privacy Principles (APPs) under the Privacy Act 1988 (Cth). Regarding LLM API usage: no confidential or sensitive information was submitted to external APIs all inputs consisted solely of publicly available academic text, aligning with the ethical use principles

outlined in the Swinburne University Research Ethics Policy. Neo4j Aura cloud instances used individual account logins and credentials were not shared in the code repository, consistent with ACSC Essential Eight cybersecurity controls.

6. Lessons Learned and Future Plan

6.1 Lessons Learned from Sprint 2

Lesson 1: Prompt Engineering Requires Systematic, Documented Evaluation

The team initially approached prompt testing informally. Adopting a structured scoring framework with explicit, weighted criteria was transformative. This aligns with Liu et al. (2023), *AI Engineering: Building Applications with Foundation Models*, which emphasises that prompt design should be treated as an engineering discipline with empirical evaluation and documented iteration cycles.

Recommendation: In Sprint 3, all prompts must be evaluated using the established scoring framework from sprint initiation. A prompt evaluation template will be added to the team's documentation library.

AI Engineering: Building Applications With Foundation Models

Chapter 5. Prompt Engineering

Prompt engineering refers to the process of crafting an instruction that gets a model to generate the desired outcome. Prompt engineering is the easiest and most common model adaptation technique. Unlike finetuning, prompt engineering guides a model's behavior without changing the model's weights. Thanks to the strong base capabilities of foundation models, many people have successfully adapted them for applications using prompt engineering alone. You should make the most out of prompting before moving to more resource-intensive techniques like finetuning.

Prompt engineering's ease of use can mislead people into thinking that there's not much to it.¹ At first glance, prompt engineering looks like it's just fiddling with words until something works. While prompt engineering indeed involves a lot of fiddling,

it also involves many interesting challenges and ingenious solutions. You can think of prompt engineering as human-to-AI communication: you communicate with AI models to get them to do what you want. Anyone can communicate, but not everyone can communicate effectively. Similarly, it's easy to write prompts but not easy to construct effective prompts.

Some people argue that "prompt engineering" lacks the rigor to qualify as an engineering discipline. However, this doesn't have to be the case. Prompt experiments should be conducted with the same rigor as any ML experiment, with systematic experimentation and evaluation.

The importance of prompt engineering is perfectly summarized by a research manager at OpenAI that I interviewed: "The problem is not with prompt engineering. It's a real and useful skill to have. The problem is when prompt engineering is the only thing

Lesson 2: Manual Extraction is Essential as a Quality Benchmark

The investment in rigorous manual extraction particularly the independent-then-merge approach proved critical to establishing a reliable quality baseline. This reinforces guidance from the LinkedIn Learning course *Prompt Engineering for LLMs* (2024): that human-generated ground truth data is indispensable for evaluating automated extraction quality in knowledge engineering contexts.

Recommendation: Maintain the independent-then-merge protocol as a non-negotiable quality standard for all future paper assignments.

Lesson 3: Schema Consistency Must Be Enforced at the Start

The Neo4j node labelling inconsistencies were a direct consequence of pairs applying their own conventions without a shared schema. This created unnecessary integration overhead and a credibility issue during the client demo.

Recommendation: Before Sprint 3 data population begins, a unified Neo4j schema document must be agreed upon and approved by the Team Leader. No data should be loaded into production graph instances without schema compliance validation.

Lesson 4: Interactive Deliverables Significantly Enhance Client Engagement

The HTML/CSS/JS comparison interface substantially elevated the quality of the Sprint Review session. Dr Tanjila's response confirmed that investing in demonstrable, interactive deliverables yields disproportionate value during client reviews, aligning with Agile principles of delivering working software (Beck et al., 2001).

Recommendation: Sprint 3 should include at least one demonstrable interactive deliverable, with the web UI expanded to incorporate the filtering feature requested during the Sprint 2 review.

Lesson 5: Early Blockers Must Be Escalated Without Delay

The prompt underperformance was identified at the end of Week 4 but not escalated until mid-Week 5, compressing the timeline for prompt finalisation.

Recommendation: Establish an explicit blocker escalation protocol — any issue unresolved within 24 hours must be raised at the next stand-up. The Team Leader will add this to the team code of conduct for Sprint 3.

6.2 Sprint 3 Future Plan – 5-Week Delivery Roadmap

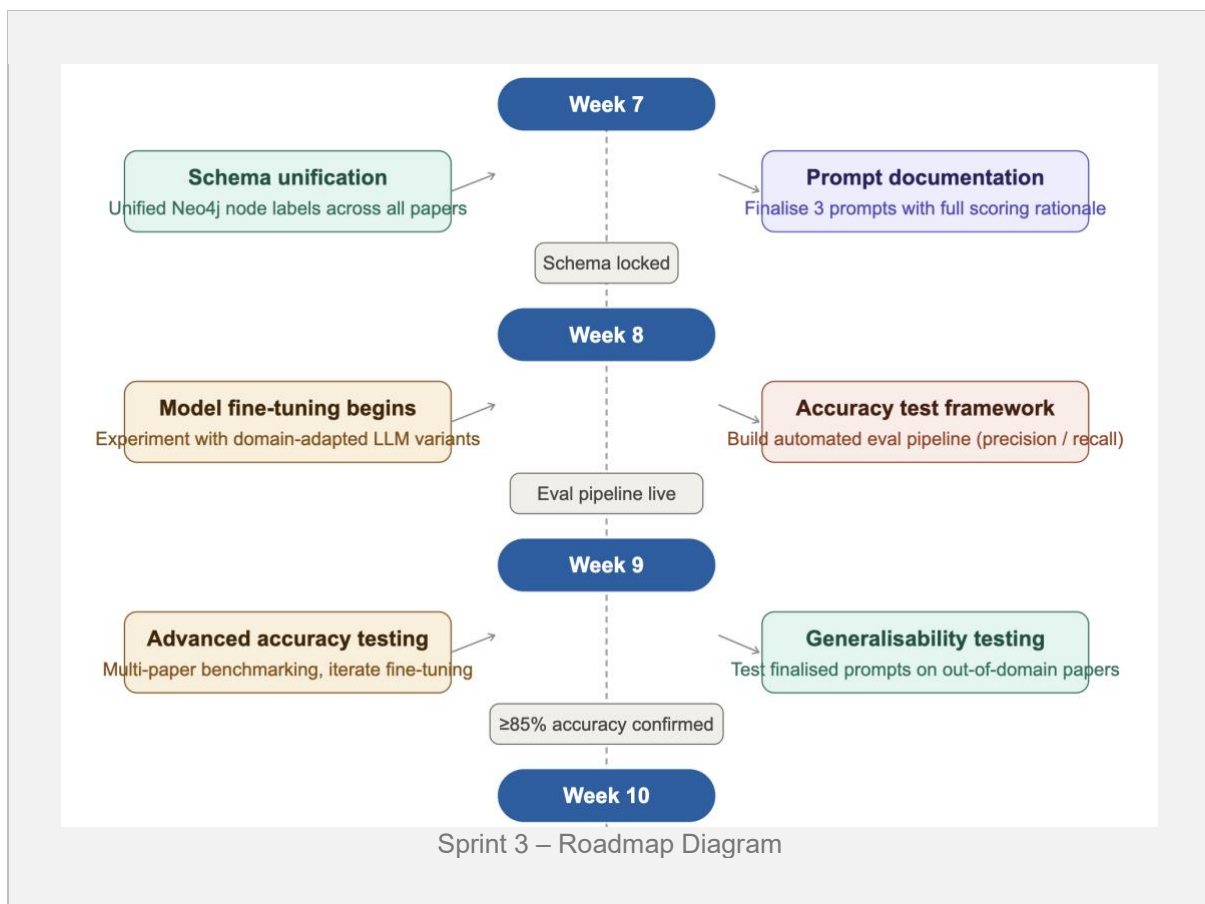
Based on the outcomes of Sprint 2 and Dr Tanjila's feedback, Sprint 3 will focus on three major technical advances: (1) fine-tuning the LLM model on domain-specific fisheries and socioeconomic data to push extraction accuracy beyond the 79% baseline achieved with prompt engineering alone; (2) finalising and formally documenting the three selected prompts with full scoring rationale; and (3) conducting more comprehensive, systematic accuracy experiments across all papers and out-of-domain test cases to build a rigorous evidence base for the final system.

The team will deliver a fully optimised, end-to-end knowledge extraction system within 5 weeks (Weeks 7–11). The following table outlines the week-by-week plan:

Week	Focus	Key Activities	Milestone	Artefact
Week 7	Schema unification & prompt documentation	Enforce unified Neo4j node labels across all papers; document final 3 prompts with full scoring rationale; add entity-type filtering to web UI	Schema locked & approved	Schema document, updated web UI
Week 8	Model fine-tuning begins & accuracy eval framework	Start domain-adapted LLM fine-tuning experiments; build automated evaluation pipeline measuring precision, recall, and F1 against manual triples	Eval pipeline live	Eval pipeline scripts, initial fine-tuning results
Week 9	Advanced accuracy testing & generalisability	Multi-paper benchmarking; iterate fine-tuning based on results; test finalised prompts against out-of-domain papers; target ≥85% accuracy	≥85% accuracy confirmed across all papers	Benchmarking report, updated prompts
Week 10	Full system integration	End-to-end pipeline: PDF → LLM → Neo4j → Web UI; integration testing;	Full pipeline operational	Integrated system, test report

Week	Focus	Key Activities	Milestone	Artefact
		performance optimisation; client progress check-in		
Week 11	Final delivery & documentation	Final accuracy validation; Sprint 3 report; client demo; complete technical documentation; handover package	Full optimised system delivered	Sprint 3 report, system documentation, demo

The roadmap visual diagram below provides a high-level overview of the 5-week delivery arc, showing the parallel and sequential nature of the planned activities:



6.3 Key References for Future Sprints

1. Liu, J., et al. (2023). AI Engineering: Building Applications with Foundation Models. O'Reilly Media. – Chapter 4 (Prompt Engineering Techniques) and Chapter 7 (Evaluation Frameworks for LLM Outputs) are directly applicable to Sprint 3 fine-tuning and accuracy testing work.
2. LinkedIn Learning. (2024). Prompt Engineering for LLMs. LinkedIn Corporation. <https://www.linkedin.com/learning> – Practical, structured course on designing and evaluating LLM prompts for real-world data extraction tasks. Recommended for all team members before Sprint 3 prompt work.

3. LinkedIn Learning. (2024). Neo4j Graph Database: Getting Started. LinkedIn Corporation. <https://www.linkedin.com/learning> – Covers advanced Neo4j schema design, Cypher query optimisation, and graph visualisation best practices.
4. White, J., et al. (2023). A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. arXiv:2302.11382. – Documents reusable prompt patterns including the Persona Pattern and Output Automator, directly applicable to the team's extraction prompts.
5. Australian Privacy Principles (APPs). Office of the Australian Information Commissioner (OAIC). <https://www.oaic.gov.au> – Required reading for all team members handling research data in Sprint 3.
6. Australian Cyber Security Centre (ACSC). Essential Eight Maturity Model. <https://www.cyber.gov.au> – Provides practical cybersecurity controls applicable to the team's cloud-hosted Neo4j instances and LLM API usage.
7. Beck, K., et al. (2001). Manifesto for Agile Software Development. <https://agilemanifesto.org> – Core reference for Agile sprint practices, particularly iterative delivery and working software as the primary measure of progress.

6.4 Sprint 3 To-Do / Recommendations

1. Develop and document a unified Neo4j schema before any data population begins
2. Apply the established prompt scoring framework from the start of all prompt engineering activities
3. Begin LLM fine-tuning experiments using domain-specific fisheries training data
4. Build and run automated accuracy evaluation pipeline (precision, recall, F1 metrics)
5. Expand the web UI to support entity-type filtering as requested by Dr Tanjila
6. Document the final three prompts with full selection rationale and scoring evidence
7. Test finalized prompts against a paper outside the fisheries domain to assess generalizability
8. Target $\geq 85\%$ accuracy across all benchmark papers before Week 9 milestone
9. Implement the 24-hour blocker escalation protocol in the team code of conduct
10. Complete LinkedIn Learning modules on Prompt Engineering and Neo4j before Week 7

References

- Australian Cyber Security Centre (ACSC). (2023). Essential Eight Maturity Model. Australian Government. <https://www.cyber.gov.au>
- Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., Jeffries, R., Kern, J., Marick, B., Martin, R. C., Mellor, S., Schwaber, K., Sutherland, J., & Thomas, D. (2001). Manifesto for Agile Software Development. <https://agilemanifesto.org>
- LinkedIn Learning. (2024). Neo4j Graph Database: Getting Started. LinkedIn Corporation. <https://www.linkedin.com/learning>
- LinkedIn Learning. (2024). Prompt Engineering for LLMs. LinkedIn Corporation. <https://www.linkedin.com/learning>
- Liu, J., Zaharia, M., & Liang, P. (2023). AI Engineering: Building Applications with Foundation Models. O'Reilly Media.
- Office of the Australian Information Commissioner (OAIC). (2019). Australian Privacy Principles. Australian Government. <https://www.oaic.gov.au>

White, J., Fu, Q., Hays, S., Sandborn, M., Olea, C., Gilbert, H., Elnashar, A., Spencer-Smith, J., & Schmidt, D. C. (2023). A prompt pattern catalog to enhance prompt engineering with ChatGPT. arXiv:2302.11382. <https://arxiv.org/abs/2302.11382>